

CASE STUDY

An exploratory analysis
of a product use case by
System Level Solutions

A Scalable and Automated
Waste Collection System

CLIENT PROFILE

Established by the Karnataka Municipal Act, the client carries out all the functions of an urban administrative division for creating a Smart E-City. The client focuses on comprehensive city development by creating physical, social, & economic infrastructure.

CHALLENGE

In vision to create a Smart E-City, the client wanted to automate and optimize its operations. The waste collection procedure was one key area which involved manual intervention leading to reduced overall efficiency. The following were the two major challenges:-

- | QR Code Maintenance
- | Manual Entry of Weight Value in Mobile App

SOLUTION

After analysing the challenges, SLS provided a weighing scale and a GPRS powered Gateway consisting of 3 in-build buttons to identify Dry, Wet and Reject waste. The data collection is done by RS232 from weighing scale to Gateway.

The solution automated the identification of measurable segregated waste weight removing the need of scanning a QR code. The weighing scale then automatically sent the waste data to a Web Dashboard for data storage, analysis and decision-making, removing the need for manual entry into the mobile application.

BENEFITS

- Reduced human intervention & error.
- Better collection, segregation & disposal of garbage.
- Increased efficiency in terms of time & effort.
- Intelligent analytical engine for informed decisions.

